

Tumor Angiogenesis as a Target for Dietary Cancer Prevention

Between 2000 and 2050, the number of new cancer patients diagnosed annually is expected to double, with an accompanying increase in treatment costs of more than \$80 billion over just the next decade. Efficacious strategies for cancer prevention will therefore be vital for improving patients' quality of life and reducing healthcare costs. Judah Folkman first proposed antiangiogenesis as a strategy for preventing dormant microtumors from progressing to invasive cancer. Although antiangiogenic drugs are now available for many advanced malignancies (colorectal, lung, breast, kidney, liver, brain, thyroid, neuroendocrine, multiple myeloma, myelodysplastic syndrome), cost and toxicity considerations preclude their broad use for cancer prevention. Potent antiangiogenic molecules have now been identified in dietary sources, suggesting that a rationally designed antiangiogenic diet could provide a safe, widely available, and novel strategy for preventing cancer. This paper presents the scientific, epidemiologic, and clinical evidence supporting the role of an antiangiogenic diet for cancer prevention.